August Ning CV

École Polytechnique Fédérale de Lausanne (EPFL)
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CH–1015 Lausanne, Switzerland

Research Interests

Computer architecture under economic and future-trend constraints; datacenter architectures and systems; chiplet architectures; cost and energy efficiency; sustainability; computing policy; chip tapeouts.

Appointments

EPFL 2025 – Present
Postdoctoral Research Associate with Prof. Babak Falsafi, PARSA, and EcoCloud.

Education

Princeton University 2020 – 2025
Ph.D., Electrical and Computer Engineering, 2025
M.A., Electrical and Computer Engineering, 2022
Advisor: Prof. David Wentzlaff
Committee: Profs. Sharad Malik, Margaret Martonosi, and Benjamin C. Lee.

Duke University 2016 – 2020
B.S.E., Electrical and Computer Engineering
B.S., Computer Science
Magna Cum Laude, Graduated with ECE Distinction
Advisor: Prof. Krishnendu Chakrabarty.

Preprints

Hengrui Zhang, Pratyush Patel, August Ning, and David Wentzlaff (2025). *SPAD: Specialized Prefill and Decode Hardware for Disaggregated LLM Inference.* arXiv: 2510.08544 [cs.AR]. URL: https://arxiv.org/abs/2510.08544

Publications

August Ning and David Wentzlaff (2025). "Chip Architectures Under Advanced Computing Sanctions". In: *Proceedings of the 52nd Annual International Symposium on Computer Architecture (ISCA).* DOI: 10.1145/3695053.3731012

Jeremiah Giordani*, Ziyang Xu*, Ella Colby, August Ning, Bhargav Reddy Godala, Ishita Chaturvedi, Shaowei Zhu, Yebin Chon, Greg Chan, Zujun Tan, Galen Collier, Jonathan D. Halverson, Enrico Armenio Deiana, Jasper Liang, Federico Sossai, Yian Su, Atmn Patel, Bangyen Pham, Nathan Greiner, Simone Campanoni, and David I. August (2024). "Revisiting Computation for Research: Practices and Trends". In: *2024 International Conference for High Performance Computing, Networking, Storage and Analysis (SC24)*. DOI: 10.1109/SC41406.2024.00076

Hengrui Zhang, August Ning, Rohan Baskar Prabhakar, and David Wentzlaff (2024). "LLM-Compass: Enabling Efficient Hardware Design for Large Language Model Inference". In: *2024 ACM/IEEE 51st Annual International Symposium on Computer Architecture (ISCA)*. DOI: 10.1109/ISCA59077.2024.00082

Ang Li, Ting-Jung Chang, Fei Gao, Tuan Ta, Georgios Tziantzioulis, Yanghui Ou, Moyang Wang, Jinzheng Tu, Kaifeng Xu, Paul Jackson, August Ning, Grigory Chirkov, Marcelo Orenes-Vera, Shady Agwa, Xiaoyu Yan, Eric Tang, Jonathan Balkind, Christopher Batten, and David Wentzlaff (2023). "CIFER: A Cache-Coherent 12nm 16mm^2 SoC With Four 64-Bit RISC-V Application Cores, 18 32-Bit RISC-V Compute Cores, and a 1541 LUT6/mm^2 Synthesizable eFPGA". in: *IEEE Solid-State Circuits Letters*. DOI: 10.1109/LSSC.2023.3303111

August Ning, Georgios Tziantzioulis, and David Wentzlaff (2023). "Supply Chain Aware Computer Architecture". In: *Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA)*. DOI: 10.1145/3579371.3589052

Ting-Jung Chang*, Ang Li*, Fei Gao, Tuan Ta, Georgios Tziantzioulis, Yanghui Ou, Moyang Wang, Jinzheng Tu, Kaifeng Xu, Paul J. Jackson, August Ning, Grigory Chirkov, Marcelo Orenes-Vera, Shady Agwa, Xiaoyu Yan, Eric Tang, Jonathan Balkind, Christopher Batten, and David Wentzlaff (2023). "CIFER: A 12nm, 16mm^2, 22-Core SoC with a 1541 LUT6/mm^2, 1.92 MOPS/LUT, Fully Synthesizable, Cache-Coherent, Embedded FPGA". in: *2023 IEEE Custom Integrated Circuits Conference (CICC)*. DOI: 10.1109/CICC57935.2023.10121294

Fei Gao, Ting-Jung Chang, Ang Li, Marcelo Orenes-Vera, Davide Giri, Paul Jackson, August Ning, Georgios Tziantzioulis, Joseph Zuckerman, Jinzheng Tu, Kaifeng Xu, Grigory Chirkov, Gabriele Tombesi, Jonathan Balkind, Margaret Martonosi, Luca Carloni, and David Wentzlaff (2023). "DECADES: A 67mm^2, 1.46TOPS, 55 Giga Cache-Coherent 64-bit RISC-V Instructions per second, Heterogeneous Manycore SoC with 109 Tiles including Accelerators, Intelligent Storage, and eFPGA in 12nm FinFET". in: *2023 IEEE Custom Integrated Circuits Conference (CICC)*. DOI: 10.1109/CICC57935.2023.10121257

Ang Li, August Ning, and David Wentzlaff (2023). "Duet: Creating Harmony between Processors and Embedded FPGAs". In: *2023 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*. DOI: 10.1109/HPCA56546.2023.10070989

Sanmitra Banerjee, Arjun Chaudhuri, August Ning, and Krishnendu Chakrabarty (2021). "Variation-Aware Delay Fault Testing for Carbon-Nanotube FET Circuits". In: *IEEE Transactions on Very Large Scale Integration (VLSI) Systems* 29.2, pp. 409–422. DOI: 10.1109/TVLSI.2020.3045417

Workshops & Presentations

Ayan Chakraborty, Ali Ansari, Yuanlong Li, Shanqing Lin, Arunkrishna AMS, Alireza Foroodnia, August Ning, Mohammad Alian, Michael Ferdman, Pejman Lofti-Kamran, and Babak Falsafi (2026). "Design Space Exploration for Optimal Silicon-Efficient Server Chiplets". In: *The 3rd Workshop on Hot Topics in System Infrastructure (HotInfra'26)*

August Ning, Xavier Ouvrard, and Babak Falsafi (2026). "Power Breakdowns and PUE Implications of Air-Cooled and Direct-Liquid-Cooled Servers". In: *The 3rd Workshop on Hot Topics in System Infrastructure (HotInfra'26)*

August Ning and David Wentzlaff (2024). "Carbon Characterization of a Megawatt-scale Research Data Center". In: *The Andlinger Center for Energy and the Environment's 2024 Annual Meeting*

August Ning and David Wentzlaff (2022). "Computer Architectures for Chip Surplus". In: *ACM Student Research Competition at MICRO 2022*

August Ning, Georgios Tziantzioulis, and David Wentzlaff (2022). "Supply Chain Aware Chip Architecture". In: *The Fourth Young Architect Workshop at ASPLOS 2022*

Grants

Swiss AI Initiative - Small Grant (45k GPU hours) — 2026
"Model-System Insights for Hardware-Efficient LLM Inference"
Co-PI; with Babak Falsafi and Sanidhya Kashyap

Talks

Sustainability Conference for Responsible Research Computing (SC4RC), "Fan, PSU, and Idle Power in Air-Cooled/Liquid-Cooled Servers"	2026
OCP EMEA Summit, "Fan, PSU, and Idle Power in Air-Cooled/Liquid-Cooled Servers"	2026
Carnegie Mellon University, Critical Technology Initiative Seminar	2026
University of Central Florida, "Architectural Implications of Advanced Computing Sanctions"	2025
EPFL, "Designing Computer Systems under Sanctions and Cost Efficiency"	2025
Princeton, ACM, "How to Apply to Grad School"	2024
UC Berkeley, SLICE Lab, "Architectural Implications of Advanced Computing Sanctions"	2024
Stanford University, Computer Architecture Reading Group, "Architectural Implications of Advanced Computing Sanctions"	2024
New York University, Computer Architecture Day, "Chip Architectures Under Advanced Computing Sanctions"	2024
Princeton, ACM, "Should You Go to Grad School?"	2024
AMD Research, "Supply Chain Aware Computer Architecture"	2023
Princeton Graduate Fellowship Panel, Panelist	2022
Princeton EGR 152, "Spotlight on Engineering"	2022

Service

Research

Program Committee: PACT 2025 SRC; ISCA 2026 (Light PC); MICRO 2026; HotCarbon 2026

Social Co-Chair, ASPLOS 2023

Artifact Evaluation Committee: ISCA 2024; HPCA 2025

Undergrad Architecture Mentoring (uArch) Workshop Mentor: ISCA 2023; MICRO 2024 (Panelist); ISCA 2025 (Panelist)

Computer Architecture Student Association (CASA) Steering Committee 2022 – 2025

SIGARCH, SIGMICRO, and TCCA

Joint Task Force on PhD Student Reviewer Training 2025

Princeton

Whitman College, Resident Graduate Student 2023 – 2025

Princeton ACM, Graduate Student Liaison 2023 – 2025

ECE Department Graduate Committee 2022 – 2024

Princeton Graduate Student Government 2021 – 2024

Duke

Duke IEEE Student Branch 2016 – 2020

Tau Beta Pi (NC Gamma), Treasurer 2019 – 2020

Engineering World Health Tanzania, Volunteer BMET Summer 2017

Teaching

EPFL

Primary Lecturer
CS 728: Topics on Datacenter Design (Spring 2026)

Princeton

Graduate Teaching Assistant
ECE/COS 475/575: Computer Architecture (Spring 2022)
Instructor: Prof. David Wentzlaff

Duke
Undergraduate Teaching Assistant
ECE 110: Fundamentals of ECE (Fall 2018; Spring 2018, 2019, 2020)
Instructor: Prof. Stacy Tantum

Undergraduate Teaching Assistant
ECE 230: Microelectronic Devices and Circuits (Fall 2018)
Instructor: Prof. Aaron Franklin

Undergraduate Teaching Assistant
ECE 350: Digital Systems (Fall 2019; Spring 2019, 2020)
Instructors: Prof. Rabih Younes; Prof. John Board

Mentoring

Computer Architecture Long-term Mentoring (CALM): Mentor (2026-27)

EPFL
Students supervised: Alireza Foroodnia (PhD), Arunkrishna AMS (PhD), Mohamad Al Shaar (MS, exchange), Luna Ralet (MS), Leila Sidjanski (MS).

Princeton
Princeton-Intel REU Program: Dara Oseyemi, Mukund Ramakrishnan, Manya Zhu (Summer 2022); Jeremy Hui, Nikhil Sampath (Summer 2024).

Research Experience

AMD Research, Research Intern — Summer 2023
Bellevue, WA
Worked with Yasuko Eckert
CPU/GPU profiling and optimization for large language model workloads.

Princeton Parallel Group — 2021 – 2025
Post-Moore's-law computer architectures
Thesis: *Computer Architecture Under Economic Constraints.*

Chakrabarty Lab — 2018 – 2020
VLSI testing for deep learning hardware and carbon nanotube FETs
Undergraduate thesis: *Automating Path Generation for Variation-Aware Delay Fault Testing.*

TU Dortmund Department of High Voltage Technology — Summer 2018
Advisor: Prof. Frank Jenau
High-voltage cable technologies and measurement systems
Supported by DAAD RISE Germany Scholarship.

Industry Experience

Microsoft, Software Engineering Intern Summer 2020
Redmond, WA (remote)
Azure hardware acceleration; project implemented in C/C++.

Burns & McDonnell, Electrical Engineering Summer Analyst Summer 2019
Chicago, IL
Substation design (physical and wiring diagrams) for ComEd and LGE-KU contracts.

Honors

Princeton, ECE Graduate Student Award for Excellence in Service	2023
Princeton, Gordon Y. S. Wu Fellowship	2020
National Science Foundation Graduate Research Fellowship	2020
Duke, Otto Meier Jr. Tau Beta Pi Award	2020
Duke, Chief Student Marshal	2019
IEEE Eta Kappa Nu	2019
Tau Beta Pi	2019
DAAD RISE Germany Scholarship	2018
Duke, Bingle Family Scholarship	2017

Travel Grants: ASPLOS 2022 (YArch), MICRO 2022 (SRC), HPCA 2023, ASPLOS 2023, ISCA 2023, ISCA 2024, ISCA 2025

Additional Information

Languages: Fluent in English and Mandarin Chinese; proficient in Spanish.
Hobbies: House plants, gardening, day hiking, cooking.
Citizenship: USA

Last updated: 18 June 2026

This CV is adapted from Sara Venkatraman's CV Templates:
https://github.com/sara-venkatraman/LaTeX-Templates